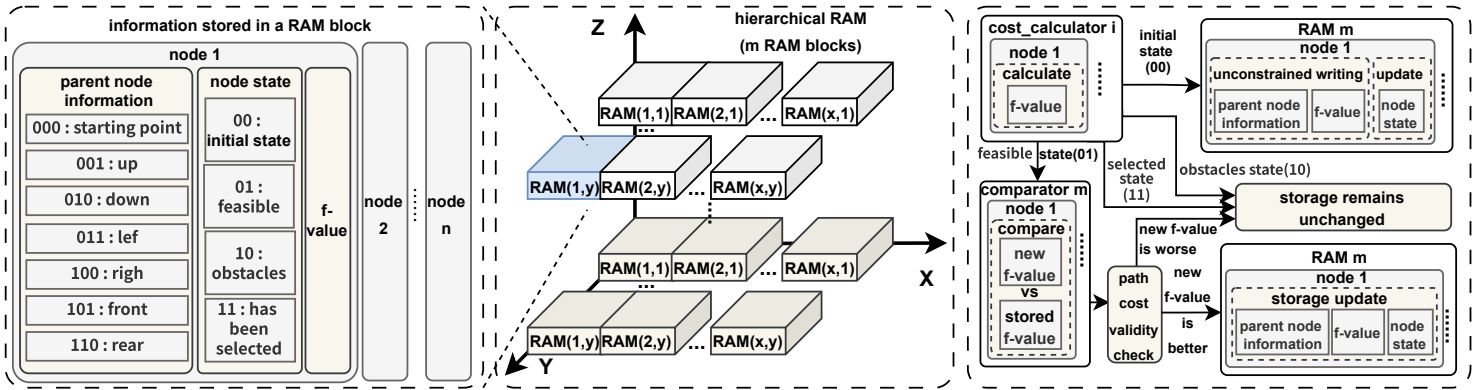

information stored in a RAM block
node 1
parent node information
000 : starting point
001 : up
010 : down
011 : lef
100 : righ
101 : front
110 : rear
node state
00 : initial state
01 : feasible
10 : obstacles
11 : has been selected
f-value
node 2
node n
(a)
hierarchical RAM
(m RAM blocks)
Z
X
Y
RAM(1,1)
RAM(2,1)
RAM(x,1)
RAM(1,y)
RAM(2,y)
RAM(x,y)
(b)
cost_calculator i
node 1
calculate
f-value
initial state (00)
RAM m
node 1
unconstrained writing
update
parent node information
f-value
node state
feasible state(01)
selected state (11)
obstacles state(10)
comparator m
node 1
compare
new f-value
vs
stored f-value
storage remains unchanged
new f-value is worse
path cost validity check
new f-value is better
RAM m
node 1
storage update
parent node information
f-value
node state
(c)